

1. JAVA Basics

1. How to create a class, object, method and its signature

Answer:

```
// Class creation
class Student {
    // Method with signature
    void displayName() {
        System.out.println("Hello Student");
    }
}

public class Main {
    public static void main(String[] args) {
        // Object creation
        Student s1 = new Student();
        // Method call
        s1.displayName();
    }
}
```

Explanation:

- **Class** → Student
- **Object** → Student s1 = new Student();
- **Method** → displayName()
- **Method Signature** → displayName() (method name + parameters)

2. Write a program to print your name.

Answer:

```
public class Main {
    public static void main(String[] args) {
        System.out.println("My name is Suman");
    }
}
```

3. Write a program for a Single line comment, multi-line and documentation comments.

Answer:

```
public class Main {
    /**
     * This is a documentation comment
     * Main method starts execution
     */
    public static void main(String[] args) {
        // This is a single-line comment
        /*
         This is a
         multi-line comment
        */
        System.out.println("Comments Example");
    }
}
```

4. Define variables for different Data Types int, Boolean, char, float, double and print on the

Console.

Answer:

```
public class Bright {
    public static void main(String[] args) {
        int age = 22;
        boolean isStudent = true;
        char grade = 'A';
        float height = 5.9f;
        double salary = 50000.75;
        System.out.println("Age: " + age);
        System.out.println("Student: " + isStudent);
        System.out.println("Grade: " + grade);
        System.out.println("Height: " + height);
        System.out.println("Salary: " + salary);
    }
}
```

5. Define the local and Global variables with the same name and print both variables and understand the scope of the variables

Answer:

```
public class Main{
    int number = 100; // Global variable (instance variable)
    public void display() {
        int number = 50; // Local variable
        System.out.println("Local Variable: " + number);
        System.out.println("Global Variable: " + this.number);
    }
    public static void main(String[] args) {
        Main obj = new Main();
        obj.display();
    }
}
```

Explanation:

- **Global variable** → Declared inside class, outside method.
- **Local variable** → Declared inside method.
- **“this.number”** is used to access the global variable.

6. Write a function to print your name and call the function from main method

Answer:

```
public class Main {
    // Function
    public void printName() {
        System.out.println("My name is Suman");
    }
    public static void main(String[] args) {
        Main obj = new Main();
        // Calling function
        obj.printName();
    }
}
```